

Calculus A

Chapter 2: Limits and Continuity

Section 2.6: Limits Involving Infinity; Asymptotes of Graphs

Main Lecture Notes

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Reference: Thomas' Calculus, 12th Edition

What We Will Look At

Fact

In this section, we will look at two main ideas:

1. Limits at infinity
2. Infinite limits
3. Horizontal, oblique, and vertical asymptotes.

Limits at ∞

Definition

$f(x)$ has limit L as x **approaches infinity** when

$$\lim_{x \to \infty} f(x) = L$$

$f(x)$ has limit L as x **approaches negative infinity** when

$$\lim_{x \to -\infty} f(x) = L$$

Note

all the limit laws (from section 2.2) apply to limits at infinity.

What This Definition Means?

Fact

In English, this definition means that as $x \to \pm\infty$, the end behavior can be determined.

End behavior means "how the function behaves as x gets further and further away from 0".

Sometimes this end behavior has the function approaching a finite value, sometimes infinity and sometimes it is undefined.

Examples

Example 1

Find $\lim_{x\to\infty} \dfrac{1}{x+1}$

We ask ourselves, "What happens as the denominator grows unbounded? (increases without bound)"

$$\lim_{x\to\infty} \frac{1}{x+1} = 0$$

Example 2

Find $\lim_{x\to-\infty} \dfrac{1}{x+1}$

Does the denominator still grow unbounded?
(decreases/increases without bound)

$$\lim_{x\to-\infty} \frac{1}{x+1} = 0$$

Rational Functions

Fact

The end behavior of rational functions can take on multiple forms: we can have

1. the function approach 0 in the long run
2. the function approach a finite number in the long run
3. the function can grow unbounded in the long run

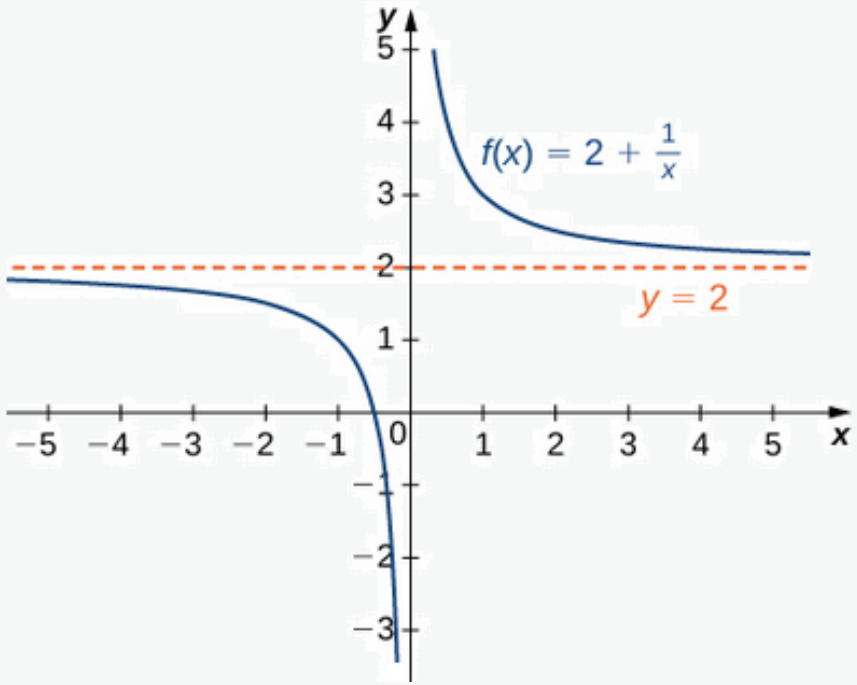
This is all based on the degrees of the numerator and denominator.

Example 3

Consider the function $f(x) = 2 + \frac{1}{x}$. As can be seen graphically in Figure and numerically in Table, as the values of x get larger, the values of $f(x)$ approach 2. Similarly, for $x < 0$, as the values $|x|$ get larger, the values of $f(x)$ approaches 2.

We have calculated the values of $f(x) = 2 + \frac{1}{x}$ for the values of x listed in Table below:

x	$2 + \frac{1}{x}$	x	$2 + \frac{1}{x}$
-10	1.9	10	2.1
-100	1.99	100	2.01
-1000	1.999	1000	2.001
-10000	1.9999	10000	2.0001



Limits at Infinity and Horizontal Asymptotes

Example 4

Determine the end behavior of $f(x) = \dfrac{x^2+1}{2x^2+3x-1}$.

Solution:

$$\lim_{x\to\infty} \frac{x^2+1}{2x^2+3x-1} = \lim_{x\to\infty} \frac{x^2+1}{2x^2+3x-1} \cdot \frac{\frac{1}{x^2}}{\frac{1}{x^2}}$$

$$= \lim_{x\to\infty} \frac{x^2\cdot\frac{1}{x^2}+1\cdot\frac{1}{x^2}}{2x^2\cdot\frac{1}{x^2}+3x\cdot\frac{1}{x^2}-1\cdot\frac{1}{x^2}}$$

$$= \lim_{x\to\infty} \frac{1+\frac{1}{x^2}}{2+\frac{3}{x}-\frac{1}{x^2}}$$

$$= \frac{1}{2}$$

Example 5

What about as $x \to -\infty$?

$$\lim_{x \to -\infty} \frac{x^2 + 1}{2x^2 + 3x - 1} = \frac{1}{2}$$

$y = \frac{1}{2}$ is called **the horizontal asymptote**.

Definition: a horizontal asymptote

A line $y = b$ is **a horizontal asymptote** of the graph of a function $y = f(x)$ if either $\lim_{x \to \infty} f(x) = b$ or $\lim_{x \to -\infty} f(x) = b$.

Example 6

Determine the end behavior of $f(x) = \frac{3x}{x^2-1}$.

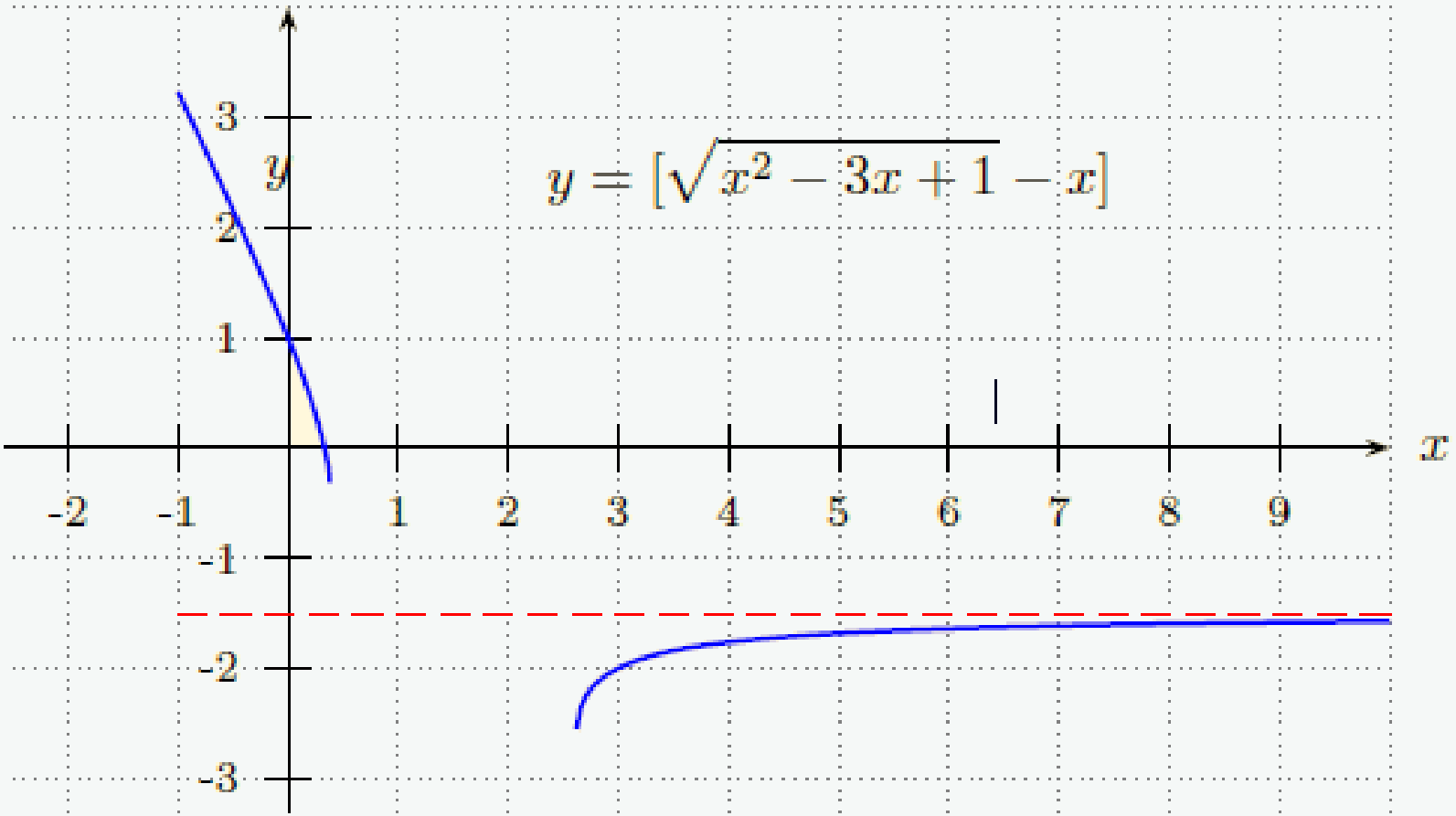
Solution:

$$\begin{aligned}
\lim_{x\to\infty} f(x) &= \lim_{x\to\infty} \frac{3x}{x^2-1} \\
&= \lim_{x\to\infty} \frac{3x}{x^2-1} \cdot \frac{\frac{1}{x^2}}{\frac{1}{x^2}} \\
&= \lim_{x\to\infty} \frac{\frac{3}{x}}{1-\frac{1}{x^2}} \\
&= 0
\end{aligned}$$

Limits at Infinity; Horizontal Asymptotes

Example 7

Determine the end behavior of $f(x) = \sqrt{x^2 - 3x + 1} - x$.



Example 7 – Solution

A direct substitution of $\lim_{x\to\infty}[\sqrt{x^2-3x+1}-x]$ gives Indeterminate Form (I.F.) $\infty-\infty$. So,

$$\lim_{x\to\infty} f(x)$$

$$= \lim_{x\to\infty}[\sqrt{x^2-3x+1}-x].\frac{\sqrt{x^2-3x+1}+x}{\sqrt{x^2-3x+1}+x}$$

$$= \lim_{x\to\infty}\frac{(x^2-3x+1)-x^2}{\sqrt{x^2-3x+1}+x} \quad \text{Simplify divide by } x$$

$$= \lim_{x\to\infty}\frac{\frac{-3x+1}{x}}{\frac{\sqrt{x^2-3x+1}+x}{x}} = \lim_{x\to\infty}\frac{-3+\frac{1}{x}}{\frac{\sqrt{x^2-3x+1}}{x}+1} \quad x\to\infty\Rightarrow x>0,\ x=\sqrt{x^2}$$

$$= \lim_{x\to\infty}\frac{-3+\frac{1}{x}}{\sqrt{\frac{x^2-3x+1}{x^2}}+1} = \lim_{x\to\infty}\frac{-3+\frac{1}{x}}{\sqrt{1-\frac{3}{x}+\frac{1}{x^2}}+1} = \frac{-3}{2}$$

The Degrees of The Polynomials

Last 2 examples:

$$\lim_{x\to-\infty}\frac{x^2+1}{2x^2+3x-1}=\frac{1}{2}$$ Example 4

$$\lim_{x\to\infty}\frac{3x}{x^2-1}=0$$ Example 6

What do we notice about the answers and the degrees of the polynomials? Can we generalize?

The Degrees of The Polynomials

1. $y = 0$ if degree of denominator is larger than degree of numerator.

2. If the degree of numerator is equal to the degree of the denominator, then the horizontal asymptote if the ratio of the lead coefficients.

What about when the degree of the denominator is smaller than that of the numerator?

3. Skew asymptote if degree of numerator is larger than degree of denominator.

Oblique Asymptote

Definition

$f(x)$ has **an oblique asymptote** if it approaches a line of the form

$$y = mx + b$$

for $m \neq 0$, as $x \to \infty$, or as $x \to -\infty$.

Note

if f is a rational function, and the degree of its numerator polynomial is 1 greater than the degree of its denominator polynomial, then f has **an oblique asymptote**.

Oblique Asymptote

Example 8

Determine the end behavior of $f(x) = \frac{x^2-1}{2x}$.

Solution:

$$\frac{x^2-1}{2x} = \frac{x}{2} - \frac{1}{2x}$$

So, as $x \to \infty$, $f(x) \to \frac{x}{2}$, $-\frac{1}{2x}$ is the remainder and goes to 0.

Infinite Limits and Vertical Asymptotes

Infinite Limits

Instead of looking at what happens as $x \to \pm\infty$, we will look at how it comes about that we have $f(x) \to \pm\infty$.

Example 9

Find $\lim\limits_{x\to 1} \dfrac{1}{(x-1)^2}$. What does this function look like?

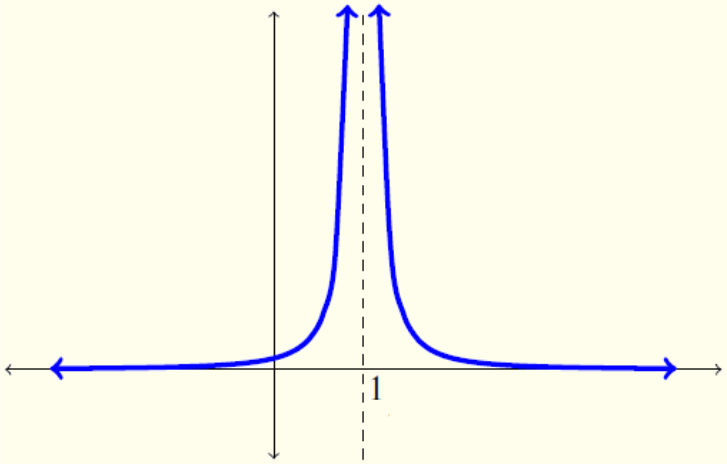
Solution:

To find the limit, we need to consider the one-sided limits as we approach $x = 1$.

$$\lim_{x\to 1^-} \frac{1}{(x-1)^2} = \frac{1}{(-\text{quantity} \to 0)^2} = +\infty$$

$$\lim_{x\to 1^+} \frac{1}{(x-1)^2} = \frac{1}{(+\text{quantity} \to 0)^2} = +\infty$$

So, we have what we call **a vertical asymptote** at $x = 1$.



Infinite Limits and Vertical Asymptotes

Vertical Asymptote

Do we have a horizontal asymptote or oblique asymptote? What is it?

Now we analytically look at the behavior as we approach the vertical asymptote.

Definition: a vertical asymptote

The line $x = a$ is **a vertical asymptote** of $y = f(x)$ if either

$$\lim_{x \to a^+} f(x) = \pm\infty, \quad \text{or} \quad \lim_{x \to a^-} f(x) = \pm\infty.$$

Clarifications about Asymptotes

Fact

Can we cross a vertical asymptote with the function?

Vertical Asymptotes

We can never cross a vertical asymptote because they represent points where a function is undefined.

Can we cross a horizontal asymptote?

Horizontal Asymptotes

We may cross horizontal asymptotes as they indicate the long-term behavior of the function but do not tell us anything about the short-term behavior.

Are asymptotes part of the graph?

Asymptotes

Asymptotes are not part of the graph, which is why they are represented by dashed lines.

Reminder Example

Example 10

Find $\lim\limits_{x\to 1} \dfrac{(x-1)^2}{x^2-1}$

Solution:

Take the limit as usual,

$$\begin{aligned}\lim_{x\to 1} \frac{(x-1)^2}{x^2-1} &= \lim_{x\to 1} \frac{(x-1)(x-1)}{(x+1)(x-1)} \\ &= \lim_{x\to 1} \frac{x-1}{x+1} \\ &= \lim_{x\to 1} \frac{x-1}{x+1} \\ &= 0\end{aligned}$$

Asymptote Example

Example 11

Determine vertical asymptotes and end behavior for $f(x) = \frac{3x^2-12}{x^2-1}$.

Solution:

First, let's look at the end behavior. What kind of limit do we need to evaluate to find the end behavior?

$$\lim_{x\to\infty} \frac{3x^2-12}{x^2-1} = 3$$

$$\lim_{x\to-\infty} \frac{3x^2-12}{x^2-1} = 3$$

So, we have a horizontal asymptote at $y = 3$.

Now, the vertical asymptotes. We need to determine where they could be first. How do we do that?

$$f(x) = \frac{3x^2-12}{x^2-1} = \frac{3(x+2)(x-2)}{(x+1)(x-1)}$$

So, where do we have vertical asymptotes? $x = \pm 1$.
How many limits do we need to evaluate to determine the behavior of the function near these values?

Asymptote Example

Example 11 Continue

$$\lim_{x\to -1} \frac{3(x+2)(x-2)}{(x+1)(x-1)}$$

We know that the value of this limit will be $\pm\infty$. Why?
Consequently, all we care about is the sign of each factor, but the individual values are meaningless.

Solution:

$$\lim_{x\to -1^-} \frac{3(x+2)(x-2)}{(x+1)(x-1)} = \lim_{x\to -1^-} \frac{(+)(+)(-)}{(-)(-)}$$

$$\Rightarrow \lim_{x\to -1^-} \frac{3(x+2)(x-2)}{(x+1)(x-1)} = -\infty \qquad \textit{Similarly},$$

$$\lim_{x\to -1^+} \frac{3(x+2)(x-2)}{(x+1)(x-1)} = \lim_{x\to -1^+} \frac{(+)(+)(-)}{(+)(-)}$$

$$\Rightarrow \lim_{x\to -1^+} \frac{3(x+2)(x-2)}{(x+1)(x-1)} = +\infty$$

So, we have a vertical asymptote at $x = -1$.

Asymptote Example

Example 11 Continue

$$\lim_{x\to 1} \frac{3(x+2)(x-2)}{(x+1)(x-1)}$$

We know that the value of this limit will be $\pm\infty$. Why?
We do the same steps as in the case of $x = 1$.

Solution:

$$\lim_{x\to 1^-} \frac{3(x+2)(x-2)}{(x+1)(x-1)} = \lim_{x\to 1^-} \frac{(+)(+)(-)}{(+)(-)}$$

$$\Rightarrow \lim_{x\to 1^-} \frac{3(x+2)(x-2)}{(x+1)(x-1)} = +\infty \qquad \textit{Similarly},$$

$$\lim_{x\to 1^+} \frac{3(x+2)(x-2)}{(x+1)(x-1)} = \lim_{x\to 1^+} \frac{(+)(+)(-)}{(+)(+)}$$

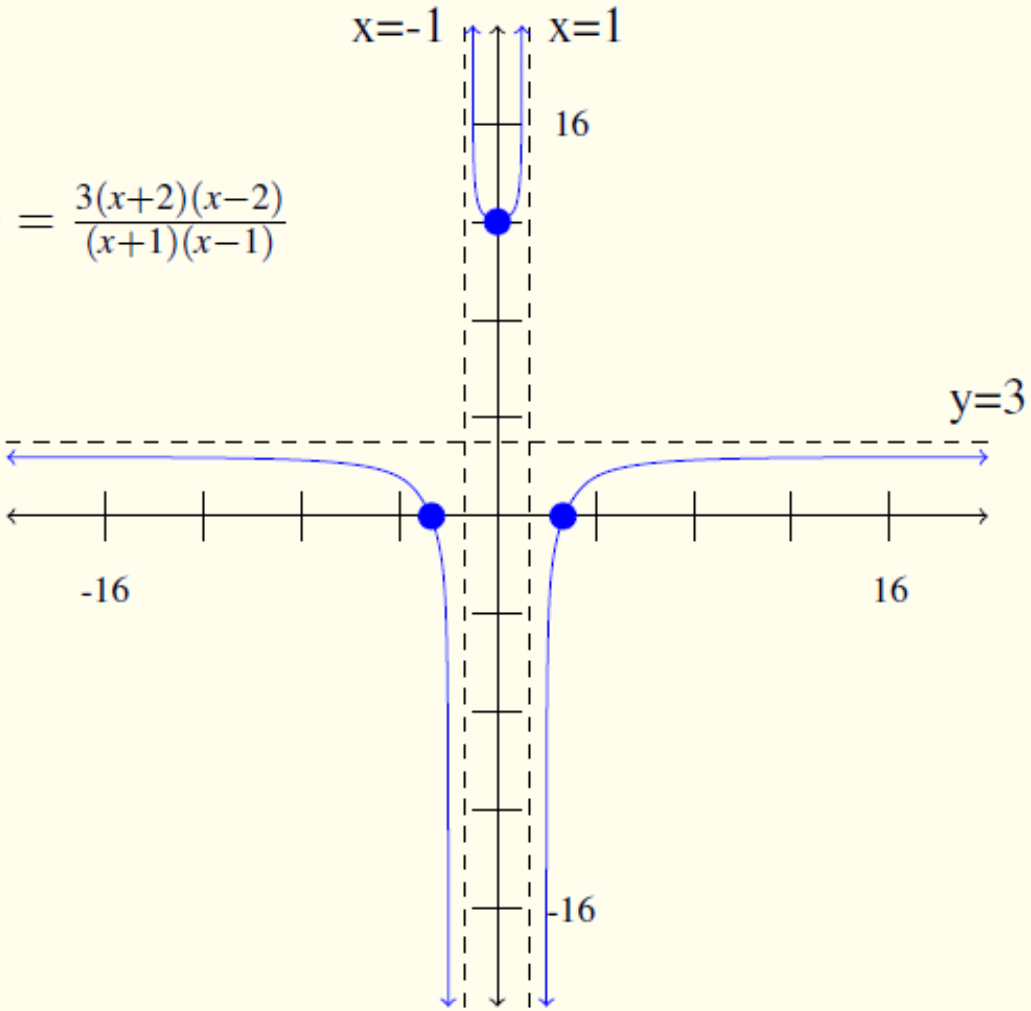
$$\Rightarrow \lim_{x\to 1^+} \frac{3(x+2)(x-2)}{(x+1)(x-1)} = -\infty$$

So, we have a vertical asymptote at $x = 1$.

Asymptote Example Plot

Solution: Example 11 Continue

$$f(x) = \frac{3(x+2)(x-2)}{(x+1)(x-1)}$$



End of Section 2.6.

Thank you.!!!